

EECE 5644

Assignment 06

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Requirement 1

The missing 1.25% of data points were interpolated using a dual methodology, using linear interpolation for small gaps (less than or equal to 60 minutes) and filling in with the same data from the previous week for larger gaps. For the smaller gaps, the assumption is that power will not change too heavily over a small timespan, so a linear interpolation is appropriate. For the larger gaps, the assumption is that daily use stays roughly consistent from one week to the next, therefore one day can be approximated with that same day from the previous week. The hourly dataset is resampled with the mean for power, voltage, and intensity, while the sub meters are summed over the hour to capture the energy used by specific appliances.

Requirement 2

Hyperparameter tuning of the model did not prove successful in achieving a MAPE of <10%. Hyperparameter tuning led to a minimum SMAPE of ~50%. SMAPE was used to better represent the differences between high and low losses, which MAPE treats the same.

$$\text{SMAPE} = \frac{2}{n} \sum_{i=1}^n \frac{|A_i - F_i|}{|A_i| + |F_i|} \times 100\%$$

Requirement 3

The train/test split was first divided with a cutoff six months before the end of the data, then frthen was windowed for the LSTM using the user defined window() function. Windowing after splitting prevents training forecasts from jumping the gap into the testing set.

Requirement 4

Running the model for a single forecast was completed in <1 ms on my machine.

```

from datetime import datetime, timedelta
with torch.no_grad():

    one_day_data = X_test[1, :, :]
    then = datetime.now()
    one_day_pred = model(one_day_data.unsqueeze(0))
    now = datetime.now()

    ttc_ms = (now - then) / timedelta(milliseconds=1)
    print(f'Time to compute [ms]: {ttc_ms}')
    assert (ttc_ms < 500) , "Time to compute greater than 500ms"

    display(one_day_pred)

```

Python

Time to compute [ms]: 0.604

```

tensor([[0.3043, 0.2831, 0.3164, 0.2877, 0.3447, 0.4361, 0.6461, 0.8338, 1.0570,
         1.2251, 1.2972, 1.2883, 1.1857, 1.0196, 0.8913, 0.9602, 1.0451, 1.0023,
         0.9049, 1.0321, 1.3604, 1.6959, 1.5520, 1.2484]], device='cuda:0')

```